



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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Objective: Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:My program is designed to simulate Incident Ticket Manager using Linear Data structure (list). It allows users to add, display, remove, and count incident tickets generated by automation bots. This shows how lists can be modified, organized, or removed.

- I. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

```
new new_node = Node(incident_id, bot, description)
```

if self.head is None:

```
self.head = new_node
```

else:

```
current = self.head
```

while current.next is not None:

```
current = current.next
```

```
current.next = new_node
```

```
self.count += 1
```

Explanation:

This lets you put a new ticket into the list when a bot reports a problem.

Display Incidents:

```
current = self.head
```

while current is not None:

```
print(f"{current.incident_id:<15}{current.bot:<20}{current.description:<35}")
```



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current = current.nextExplanation: This shows you every ticket that is still active.

Search Incident:

current = self.head

while current is not None:

if current.incident_id == incident_id:

return current

current = current.nextExplanation: This lets you look for one specific id by typing its incident ID.

Remove Resolved Incident:

if current.incident_id == incident_id:

if previous is None:

self.head = current.next

else:

previous.next = current.next

self.Count -= 1

Explanation: This takes a ticket out of the list once the problem has been fixed.



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Count Active Incidents:

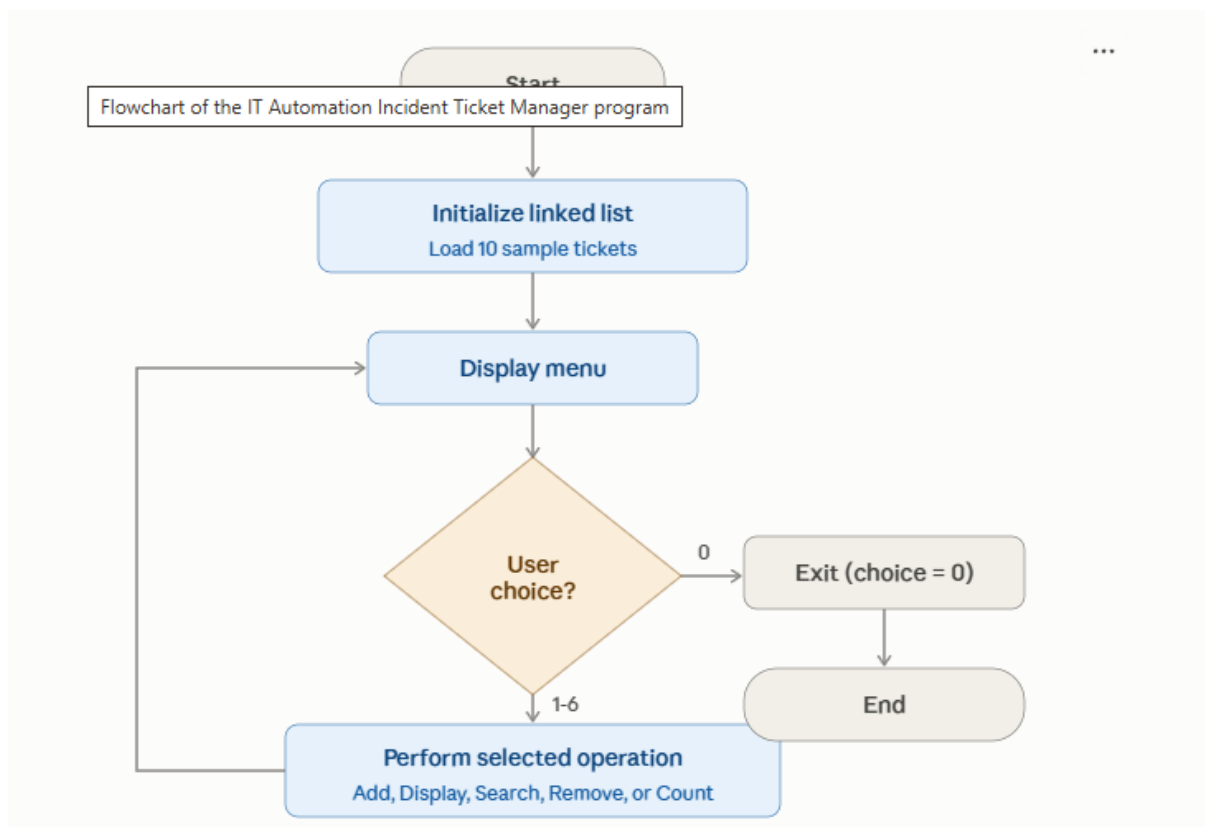
```
self.count += 1 # in add_ticket
```

```
self.count -= 1 # in remove_ticket
```

Explanation: This tells how many tickets are still active right now.

```
print(f"[i] Total active incident tickets: {self.count}\n")
```

- II. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



- III. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

Doing this activity really helped me understand what a linked list is, instead of just knowing the definition. Before this, I always thought of a "list" as just Python's built-in list, but building my own Node class and manually connecting each ticket with .next made me realize a linked list isn't stored together in memory like an array it's more like a chain of separate boxes, where each box just knows where the next one is. That's why you don't need to "shift" anything when adding or removing items, unlike arrays.